DSP – MT Exam- Summer 26- All Laws - 20 degrees

26/ 7/ 2026- (60m)

Question 1 [10 Points] [Los: 1]

a- In detail, what are the main characteristics of the discrete time sinusoidal signals? 2.5

b- Check the periodicity of the following signal, $x_a(t) = 3\cos(5t + \pi/6)$ 1.5

c- For the signal $x(t) = \cos(200\pi t)$ to be sampled with sampling frequency $F_S = 400$Hz.

(i) Find an expression for its corresponding sequence, x(n). Then calculate its fundamental period N_P. 3

(ii) Find the samples values of the first cycle in x(n), obtained in part (i), then plot x(n) and $x(t)$ at the same figure. 3

Question 2 [10 Points] [Los: 1, 3]

a- In sampling process, drive the sampling condition. 1.5

b- For the following sequence: $x(n) = \{1, 0, 0, 1\}$:

(i) Using FFT algorithm find the frequency components of x(n), X(k). 2

(ii) Check your results, obtained in part (i), by calculating x(n) using IDFT method. 2

(iii) Using your results, verify Parseval's theorem. 1.5

(iv) Calculate and plot the amplitude and the phase spectrums. 1.5

(v) What is the computational saving when using FFT algorithm instead of DFT algorithm? 1.5

GOOD LUCK